

Photosynthesis in Higher Plants

Photosynthesis is critical for two main reasons: it is the primary source of all food on Earth, and it is responsible for the release of oxygen into the atmosphere by green plants.

The study of physiological processes like photosynthesis, respiration, and plant growth is described in molecular terms, yet within the context of cellular activities and the organism level, relating these processes to the environment when appropriate.

II. Early Experiments and Understanding

Basic experiments demonstrate that chlorophyll (green pigment), light, and CO_2 are required for photosynthesis.

- Requirements: Experiments using variegated or partially covered leaves showed that photosynthesis (indicated by starch formation) only occurred in the green parts in the presence of light. An experiment using soaked cotton (which absorbs) showed that the part of the leaf exposed to air tested positive for starch, while the enclosed part tested negative, confirming that CO_2 is required.
- Joseph Priestley (1770): Revealed the essential role of air. He observed that a burning candle or a breathing mouse in a closed bell jar would "damage the air," but placing a mint plant in the jar restored the air, allowing the candle to burn and the mouse to live. Priestley discovered oxygen in 1774.
- Jan Ingenhousz (1730-1799): Using a similar setup, he showed that sunlight is essential for the plant process that purifies the air. In an elegant experiment using an aquatic

purifies the air. In an elegant experiment using an aquatic plant, he found that small bubbles () were formed around the green parts only in bright sunlight, confirming that only green parts release oxygen.

- Julius von Sachs (c. 1854): Provided evidence that plants produce glucose, which is typically stored as starch. He showed that the green substance (chlorophyll) is located in special bodies (chloroplasts) and that glucose is synthesized in the green parts of the plant.

- T.W. Engelmann (1843-1909): Using a prism, he split light and illuminated the alga *Cladophora* placed in a suspension of aerobic bacteria. The bacteria accumulated mainly in the blue and red light regions, producing the first action spectrum of photosynthesis.

- Cornelius van Niel (1897-1985): Based on studies of purple and green bacteria, he demonstrated that photosynthesis is a light-dependent reaction where hydrogen from a suitable oxidisable compound reduces to carbohydrates. He inferred that the evolved by green plants comes from H_2O , not from , a conclusion later confirmed by radioisotopic techniques.

- Melvin Calvin (1911): Used radioactive in algal photosynthesis studies just after World War II to study the formation of sugar. He mapped the pathway of carbon assimilation, proposing that plants convert light energy to chemical energy by transferring an electron in an array of pigment molecules. This work earned him the Nobel Prize in 1961. He identified the first fixation product as 3-phosphoglyceric acid (PGA), a 3-carbon organic acid.

The currently accepted overall equation for photosynthesis is:

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III. The Site of Photosynthesis

Photosynthesis takes place primarily in the green leaves and other green parts of the plant.

- **Chloroplasts:** The mesophyll cells in leaves contain a large number of chloroplasts. Chloroplasts align themselves along the walls of mesophyll cells to achieve the optimum quantity of incident light.

- **Division of Labour in Chloroplasts:**

- **Membrane System (Grana and Stroma Lamellae):** This system traps light energy and synthesizes the high-energy chemical intermediates ATP and NADPH. These are called light reactions (or photochemical reactions) as they are directly light driven.

- **Stroma (Matrix):** Enzymatic reactions occur here to synthesize sugar, which then forms starch. These are called dark reactions (or carbon reactions), although they are not directly light driven, they are dependent on the products of the light reactions (ATP and NADPH).

IV. Photosynthetic Pigments and Light Reaction

The green colour observed in leaves is due to four pigments, separable by paper chromatography: Chlorophyll a (bright or blue green), Chlorophyll b (yellow green), Xanthophylls (yellow), and Carotenoids (yellow to yellow-orange).

- Chlorophyll a is the chief pigment associated with

- The highest rate of photosynthesis occurs at the wavelengths where chlorophyll a shows maximum absorption: the blue and red regions of the spectrum.
- Accessory Pigments (chlorophyll b, xanthophylls, and carotenoids) absorb light and transfer the energy to chlorophyll a. They enhance efficiency by utilizing a wider range of incoming light wavelengths and protect chlorophyll a from photo-oxidation.

A. The Photochemical Phase (Light Reaction)

The light reaction includes light absorption, water splitting, oxygen release, and the formation of ATP and NADPH.

- Photosystems (PS I and PS II): Pigments are organized into discrete Light Harvesting Complexes (LHC), also called antennae. Each LHC funnels energy to a single chlorophyll a molecule, which forms the reaction centre.
 - PS I reaction centre absorbs maximally at 700nm
 - PS II reaction centre absorbs maximally at 680nm

B. Electron Transport (Z-Scheme)

1. PS II: absorbs red light, exciting electrons to a higher orbit. These electrons are accepted and passed through an electron transport system (ETS) containing cytochromes. This movement is "downhill" in terms of redox potential.

2. PS I: Electrons are passed to PS I. PS I absorbs red light, exciting its electrons, which are transferred to another acceptor.

3. **NADP⁺ Reduction:** These electrons move downhill again to a molecule of NADP⁺, reducing it to NADPH + H⁺.

4. **The Z Scheme:** This entire transfer scheme, from PS II, down the ETS to PS I, and finally down to , is known as the Z scheme due to its characteristic shape when carriers are placed on a redox potential scale.

C. Splitting of Water

The electrons removed from PS II are continuously replaced by the splitting of water.

- The water splitting complex is associated with PS II and is located on the inner side of the thylakoid membrane.
- Water is split into protons (H⁺), oxygen (O₂), and electrons.
- The splitting of water releases O₂ (a net product) and accumulates into the lumen of the thylakoid.

D. Photo-phosphorylation

This is the synthesis of ATP from and inorganic phosphate (P_i) in the presence of light.

- **Non-cyclic Photo-phosphorylation:** Occurs when both PS II and PS I work in series (Z scheme). This flow synthesizes both ATP and NADPH + H⁺.
- **Cyclic Photo-phosphorylation:** Occurs when only PS I is functional. It takes place in the stroma lamellae, which lack PS II and the reductase enzyme. The excited electron cycles back to the PS I complex, resulting only in the synthesis of ATP. This process also occurs when light wavelengths beyond

The excited electron cycles back to the PS I complex, resulting only in the synthesis of ATP. This process also occurs when light wavelengths beyond are available.

E. Chemiosmotic Hypothesis (ATP Synthesis)

ATP synthesis is linked to the development of a proton gradient across the thylakoid membrane, with protons accumulating towards the lumen (inside).

The development of the proton gradient is caused by three key events:

1. Water splitting occurs inside the membrane (lumen), releasing there.
2. Proton transport occurs during electron flow; carriers pick up a proton from the stroma and release it into the lumen.
3. NADPH⁺ reduction occurs on the stroma side using the reductase enzyme, which consumes from the stroma.

This process results in a decrease of protons in the stroma and a significant accumulation in the lumen, creating a proton gradient and a measurable decrease in in the lumen.

- ATP Synthase: The breakdown of this gradient provides the energy for ATP synthesis. Protons flow back to the stroma through the transmembrane channel (C) of the ATP synthase enzyme.

- The part is embedded in the membrane for proton facilitated diffusion. The part protrudes on the stroma-facing surface. The energy released from the gradient breakdown causes a conformational change in σ , catalyzing the synthesis

- Chemiosmosis requires a membrane, a proton pump, a proton gradient, and ATP synthase.

V. The Biosynthetic Phase (Dark Reaction)

The products of the light reaction (and) are used immediately in the stroma to drive the synthesis of sugars. This biosynthetic phase is not directly light-dependent but relies on the products of the light reaction, , and . The process continues for some time even after light becomes unavailable.

A. The C₃ Pathway (Calvin Cycle)

The first product of fixation identified was the 3-carbon acid, 3-phosphoglyceric acid (PGA). Plants where the first stable product is a acid follow the pathway.

- The pathway occurs in all photosynthetic plants.
- The primary acceptor is the 5-carbon ketose sugar, ribulose biphosphate (RuBP). The cycle is essential for regenerating to continue uninterrupted.

Stages of the Calvin Cycle:

1. Carboxylation: The most crucial step where is fixed into to form two molecules of 3-PGA. This reaction is catalyzed by the enzyme RuBP carboxylase-oxygenase {RuBisCO}.
2. Reduction: A series of reactions leading to glucose formation, utilizing 2 for phosphorylation and 2 for reduction per molecule fixed.
3. Regeneration: is regenerated, requiring 1 for

Energy Requirements:

- Per molecule fixed, 3 ATP and 2 {NADPH} are required.
- The cycle must turn six times to produce one molecule of glucose, requiring a total of 18 and 12 . The need for more than is potentially met by cyclic phosphorylation.

B. The C₄ Pathway (Hatch and Slack Pathway)

plants are adapted to dry tropical regions. Although they initially fix into a acid, they rely on the Calvin cycle for final sugar synthesis.

Characteristics of text C₄ Plants:

- Special leaf anatomy (Kranz anatomy).
- High tolerance to higher temperatures.
- Response to high light intensities.
- Lack photorespiration.
- Greater productivity of biomass.

Kranz Anatomy: This anatomy is characterized by large bundle sheath cells arranged in a "wreath" (Kranz) around the vascular bundles. These cells have a large number of chloroplasts, thick walls impervious to gaseous exchange, and no intercellular spaces.

C₄ Fixation Steps:

1. Primary CO₂ Acceptor: The 3-carbon molecule phosphoenol pyruvate (PEP), located in the mesophyll cells.

2. Enzyme: The fixation is catalyzed by PEP carboxylase PEPcase, which is present in mesophyll cells, where RuBisCO is absent.

3. First Product: The 4-carbon acid oxaloacetic acid (OAA) is formed in the mesophyll cells.

4. acids (OAA, malic acid, or aspartic acid) are transported to the bundle sheath cells.

5. In the bundle sheath cells, acids break down to release and a 3-carbon molecule.

6. The released enters the Calvin pathway, which takes place in the bundle sheath cells. The bundle sheath cells are rich in RuBisCO but lack .

7. The 3-carbon molecule returns to the mesophyll to regenerate PEP, completing the cycle.

Crucially, in plants, the Calvin pathway occurs only in the bundle sheath cells, whereas in plants, it occurs in all mesophyll cells.

VI. Photorespiration

Photorespiration is a process that significantly differentiates and plants.

- RuBisCO Activity: is the most abundant enzyme in the world. Its active site can bind to both and competitively, with the binding determined by the relative concentration of the two gases.

- In C3 Plants: Under high conditions, binds to , decreasing

- **Wasteful Process:** The photorespiratory pathway is wasteful because it results in the release of with the utilization of ATP, but produces neither sugars nor ATP/NADPH. Its biological function is unknown.
- **In C₄ Plants:** Photorespiration does not occur. This is because the mechanism effectively releases inside the bundle sheath cells, increasing the concentration at the site of . This ensures that primarily functions as a carboxylase, minimizing its oxygenase activity. The resulting lack of photorespiration contributes to the higher productivity and yield of plants.

VII. Factors Affecting Photosynthesis

The rate of photosynthesis affects the yield of crop plants. It is controlled by both internal (plant) and external factors.

- **Internal (Plant) Factors:** These factors are dependent on the plant's genetic predisposition and growth. They include the number, size, age, and orientation of leaves; the state of mesophyll cells and chloroplasts; internal concentration; and the amount of chlorophyll.
- **External Factors:** These include sunlight availability, temperature, concentration, and water.

When multiple factors affect a process, Blackman's (1905) Law of Limiting Factors applies: the rate is determined by the factor that is nearest to its minimal value, and a change in this factor directly affects the rate.

A. Light

- At low light intensities, the rate of fixation shows a linear relationship with incident light.
- At high light intensities, the rate gradually becomes saturated as other factors become limiting.
- Light saturation occurs at only about 10 per cent of full sunlight. Therefore, light is rarely a limiting factor in nature, except for shade plants.
- Incident light beyond a certain point can cause the breakdown of chlorophyll and decrease photosynthesis.

B. Carbon Dioxide Concentration

is the major limiting factor for photosynthesis.

- Current atmospheric concentration is very low (0.03 to 0.04 per cent). Increasing concentration up to 0.05 per cent can increase fixation rates, but beyond that, it can become damaging.
- Response Differences: and plants respond differently to high under high light conditions.
 - plants show saturation at about .
 - plants continue to increase their rate and only saturate beyond .
- Current availability is limiting to plants. Greenhouse crops like tomatoes and bell pepper are grown in -enriched atmospheres to achieve higher yields, leveraging the response to higher concentration.

C. Temperature

The dark reactions are highly temperature controlled because they are enzymatic.

- plants respond to higher temperatures and show a higher rate of photosynthesis, while plants have a much lower temperature optimum.
- The temperature optimum for any plant depends on the habitat to which it is adapted; tropical plants have a higher optimum than temperate plants.

D. Water

The effect of water is mostly indirect, affecting the plant rather than directly impacting the chemical reaction. Water stress causes the stomata to close, thereby reducing availability. Water stress also leads to wilting, which reduces the leaf surface area and metabolic activity